



FULL PRODUCTION PRODUCT BIBLE · V3.0 · CONFIDENTIAL

The Complete CCUS Engineering Platform.

The full production vision for CarbonLens: from CO₂ storage studio to complete CCUS engineering platform — covering Capture, Transport, Utilization and Storage, with enterprise security, global regulatory frameworks, LAS/GIS data integration, team collaboration, academic partnership programme, and a clear 36-month path to \$10M ARR.

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Full Platform Vision

THE 36-MONTH ARC

The PoC validates CO₂ storage simulation as a commercial product. The Full Production Platform extends that validated core into a complete CCUS engineering environment — the first browser-based tool to cover the entire carbon value chain from source characterisation through long-term monitoring. This document is the blueprint for what is built after the PoC validates market demand.

YEAR 1 — MONTHS 1-13	YEAR 2 — MONTHS 14-24	YEAR 3 — MONTHS 25-36
PoC → Storage Studio Revenue Execute the PoC plan. CO₂ Storage Studio fully featured. \$10K MRR. Provisional patent filed. Design-partner customers co-designing features for real permit applications. Pre-seed fundraising initiated.	Full Platform Build Add Capture + Transport + Utilization modules. Full MMV planner. GIS import. 8 regulatory jurisdictions. Team collaboration. Enterprise security (SOC 2 preparation). Academic partnership programme active. \$150K–\$360K ARR.	Platform → Market Leader Subsurface AI assistant. Full API layer extracted from GUI. Enterprise deals with major consultancies and project operators. \$1M–\$3.6M ARR. Decision point: raise Series A to accelerate, or accept strategic acquisition from CMG/SLB at \$5–15M.

Platform Positioning — What CarbonLens Becomes

The Full Production Platform positions CarbonLens as: **“The operating system for CCUS project development — from source to storage, from feasibility to permit, from permit to monitoring.”** This is a platform play, not a product play. Every CCUS project team needs what CarbonLens provides. The network effect is the formation library: as more projects publish results using CarbonLens, the preset library grows, validating the tool for the next project team.

What Separates Full Production from PoC

DIMENSION	POC (YEAR 1)	FULL PRODUCTION (YEAR 2-3)
CCUS Coverage	Storage only (honest scope)	Full chain: Capture + Transport + Utilization + Storage + MMV
Regulatory	4 jurisdictions (EPA, UK, Norway, PETRONAS)	8+ jurisdictions including Australia, Canada, Saudi, EU ETS
Data Integration	LAS 2.0 import	LAS + GIS shapefiles + multi-well datasets + subsurface API integrations
Collaboration	Single-user cloud save	Team workspaces, version history, client report templates, audit logs

Security	Cloudflare auth, HTTPS	SOC 2 Type II, enterprise SSO (SAML), data residency options, pen testing
ML Models	5 core models (IFT, density, viscosity, solubility, site scorer)	+MMP predictor, impurity effects, reservoir heterogeneity ML, AI assistant
Onboarding	Basic guided tutorial + presets	Role-based onboarding, video library, SPE-accredited CPD course integration
Revenue	\$10K MRR target (Month 13)	\$100K+ MRR target (Month 36)

Complete Module Architecture

The Full Production Platform consists of nine integrated modules. Modules 1–5 are the PoC-validated core (Storage Studio). Modules 6–8 complete the CCUS value chain. Module 9 is the AI-powered synthesis layer that ties the platform together.

MODULE 1 – CORE

Reservoir Builder & Simulation Studio

Parametric 3D/2D reservoir geometry, well placement, real-time CO₂ plume simulation, LAS import, formation library. Validated in PoC.

MODULE 2 – CORE

ML Property Engine

All 7 CO₂-brine thermophysical properties with P10/P50/P90 bounds. Hybrid client-side/server-side architecture. Validated in PoC.

MODULE 3 – CORE

Uncertainty Quantification Engine

Monte Carlo n=200/1,000. P10/P50/P90 plume envelopes. AoR risk chart. Validated in PoC.

MODULE 4 – CORE

Geomechanical & Permit Suite

FPG, Eaton stress, MAIP, caprock seal index. Permit export for 8 jurisdictions. Added in PoC P1.

MODULE 5 – CORE

Site Screening & Suitability Scorer

ML-based 0–100 suitability score. NETL SRMS capacity method. RAG status display. Validated in PoC.

MODULE 6 – YEAR 2

Capture + Transport + Utilization

Source characterisation, capture technology selector and cost estimator, pipeline sizing calculator, CO₂-EOR MMP predictor, utilization pathway selector.

MODULE 7 – YEAR 2

Full MMV Planner

Monitoring well location optimiser, pressure monitoring schedule generator, USDW protection zone delineation, seismic survey planning, compliance reporting automation.

MODULE 8 – YEAR 2

Team Collaboration Workspace

Named project library, version history, team sharing with role-based permissions, client-branded report generation, export audit logs for regulatory submissions.

MODULE 9 – YEAR 3

Subsurface AI Assistant

Natural language interface: “What injection rate maximises dissolution trapping for this formation?” Powered by Claude API with CarbonLens tool-calling. Retrieval-augmented generation from the CCUS literature corpus.

Full CCUS Coverage & Data Integration

03 — Complete CCUS Value Chain (Year 2)

The storage studio validates commercial viability. The full CCUS platform earns the name. Each upstream module adds a distinct ICP and expands addressable revenue. These modules do not require new ML model development — they use literature correlations and decision-tree calculators, making them relatively fast to implement compared to the storage core.

MODULE	KEY FEATURES	NEW ICP UNLOCKED	BUILD EFFORT
Capture Module	CO ₂ purity / flow rate / impurity characterisation. Capture technology selector (post-combustion, oxy-fuel, DAC, pre-combustion). Capital + OPEX cost estimator benchmarked against IEA/NETL data. Purity impact on IFT (feeds storage module).	Industrial emitters (power, cement, steel) needing capture technology evaluation	Medium — literature correlations, no new ML training
Transport Module	Pipeline sizing calculator (Darcy-Weisbach for dense-phase CO ₂). Phase behaviour during transport (supercritical vs. dense-phase boundary). Pressure drop over distance. Pump/compressor sizing. Impurity effects on transport properties.	Pipeline operators, infrastructure developers, project economists	Low-Medium — well-established engineering correlations
Utilization Module	CO ₂ -EOR Minimum Miscibility Pressure (MMP) predictor (ML model from petroleum engineering literature). Utilization pathway selector: EOR, mineralisation, food-grade, synthetic fuels, concrete curing. 45Q utilization vs. storage credit comparison.	EOR operators, utilization project developers — expands market beyond pure storage	Medium — MMP is a specific ML model to train (already in v1 roadmap)

04 — Data Integration Architecture

The PoC adds LAS 2.0 import. The full platform extends data integration to cover all common formats that real subsurface engineers use, making CarbonLens the single platform where all their project data lives.

DATA FORMAT	WHAT IT CONTAINS	CARBONLENS USE	PRIORITY
LAS 2.0 / 3.0	Well log curves (porosity, density, resistivity, temperature)	Auto-populate formation parameters in reservoir builder	PoC P1 ✓

CSV / Excel	Formation parameter tables from geological surveys	Bulk formation data import, batch site screening	Year 1 post-PoC
GIS Shapefiles (.shp)	Basin geometry, fault maps, formation boundaries	Auto-generate reservoir geometry from real basin shape data	Year 2
SEGY (seismic)	3D seismic cube data	Display seismic cross-section alongside CarbonLens simulation (view-only, no processing)	Year 2–3
ECLIPSE / CMG output	Full simulation results (pressure, saturation)	Import competitor simulation results for comparison against CarbonLens outputs	Year 2

Global Regulatory Framework & Team Collaboration

05 — Global Regulatory Framework (8 Jurisdictions)

JURISDICTION	REGULATOR	MARKET	CARBONLENS TEMPLATE	TIMELINE
USA — EPA Class VI	US EPA UIC Programme	232+ active apps. Primary. \$43.4B 45Q.	Full permit suite. MAIP, AoR, MVA plan. PoC ✓	PoC
UK — NSTA	North Sea Transition Authority	Drax, BP Teesside, HyNet. Billions committed.	SP application, plume footprint, NSTA reporting format	PoC P1
Norway — NPD	Norwegian Petroleum Directorate	Northern Lights, Equinor. World benchmark.	Petroleum Act storage permit format, geomechanical requirements	PoC P1
Malaysia — PETRONAS	PETRONAS / MNR	First-mover. No tool exists. UTP access.	PETRONAS CCS feasibility template. Offshore format.	PoC P1
Australia — NOPSEMA	National Offshore Petroleum Safety & Environmental Management Authority	Otway Basin, CarbonNet, Gorgon.	GHGSRS permit template, NOPSEMA environment plan	Year 2
Canada — AER	Alberta Energy Regulator	Quest CCS, Weyburn. Strong pipeline.	AER Directive 51 storage well application format	Year 2
Saudi Arabia	ARAMCO internal + GAMEP	ARAMCO CCS master plan. Large strategic market.	ARAMCO project summary format + GAMEP environmental disclosure	Year 2–3
EU ETS — Annex I	European Commission	EU Innovation Fund CCS	EU ETS monitoring plan, CCS Directive storage permit	Year 2–3

06 – Team Collaboration Workspace (Studio+ Tier)

Single-user cloud save satisfies the PoC. Enterprise consultancy and project operator customers need full team workflows. This is what justifies the \$1,500/month Studio tier and the \$25K+ Enterprise contracts.

- **Named project library:** Cloudflare D1 (SQLite at edge) stores project metadata, thumbnails, last-modified timestamps. Team members see all shared projects in a sortable library.
- **Version history:** Append-only KV entries per project. Roll back to any previous saved state. “Snapshots” allow engineers to compare MAIP at different stages of a project.
- **Role-based permissions:** Owner / Editor / Viewer. Viewer access for clients to see simulation results without edit access. Invitation via email link.
- **Client report generation:** Branded PDF reports auto-generated from simulation results. Client logo and firm name appear in report header. One-click delivery via shareable link or email from within the platform.
- **Audit logs:** Every parameter change, simulation run, and export is timestamped and attributed. Required for regulatory submissions where provenance of calculations must be demonstrated.

Security Architecture & IP Strategy

07 — Security Architecture by Tier

POC — YEAR 1 Baseline Security HTTPS everywhere (Cloudflare TLS) JWT auth via Cloudflare Workers Stripe handling all payment data (PCI DSS compliant) No PII stored beyond email + Stripe ID Project data encrypted at rest in Cloudflare KV Rate limiting on Cloudflare Worker ensemble endpoint Input validation on all formation parameters (prevent injection)	YEAR 2 — ENTERPRISE READY Enterprise Security Layer SAML 2.0 SSO integration (Okta, Azure AD) Data residency options (EU, US, APAC Cloudflare regions) Field-level encryption for sensitive formation data Penetration testing (third-party, annual) GDPR / PDPA (Malaysia) compliance documentation Vulnerability disclosure programme Security questionnaire response library for enterprise procurement
YEAR 3 — SOC 2 SOC 2 Type II Certification SOC 2 Type I audit (6-month evidence collection) SOC 2 Type II audit (12-month observation period) Security, Availability, Confidentiality trust service criteria Annual recertification SOC 2 report available under NDA to enterprise prospects This unlocks Fortune 500 and government procurement	IP SECURITY — ALL TIERS ML Model Protection Ensemble weights server-side (Cloudflare Worker, never in bundle) Classical equations client-side (published literature, no IP risk) JavaScript bundle minification + obfuscation Rate limiting prevents systematic equation extraction via API UTP lab dataset as trade secret (never published raw) Patent portfolio covers ensemble method and architecture

08 — IP Strategy: Patent Portfolio Roadmap

FILING	CLAIM	TIMELINE	JURISDICTION
Provisional #1	GMDH+MGGP+MARS ensemble method with bootstrap P10/P50/P90 for CO ₂ -brine thermophysical properties + hybrid client-side/server-side architecture for browser-based reservoir simulation	Week 7 of PoC	USPTO (micro-entity)
Full Patent #1	Convert provisional to full utility patent with expanded claims	Month 15–18 (from pre-seed proceeds)	USPTO + PCT (international)
Provisional #2	UQ-to-permit translation method: automated generation of EPA Class VI AoR/MAIP from ML confidence intervals	Month 10–12	USPTO

Provisional #3	Browser-native CO ₂ storage MMV planner with real-time AoR evolution visualisation	Month 18–20 (post-MMV module build)	USPTO
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COMPETITIVE MOAT – THE COMPOUNDING ADVANTAGE

The hybrid IP architecture + proprietary UTP lab data + growing formation library creates a compounding moat. Each real project that uses CarbonLens adds to the formation library. Each academic paper citing it validates the ML accuracy. Each jurisdiction template added blocks a competitor from matching the regulatory breadth. Time advantage compounds: every month of first-mover status is a month of user data, citations, and market trust that a late entrant cannot buy.

Academic Programme & Full Financial Vision

09 — Academic Partnership Programme

THE ACADEMIC CHANNEL IS UNDERVALUED IN BOTH V1 AND V2

The v2 roadmap jumps straight to cold emails to EPA applicants. For higher institution adoption, the path is different and must be explicit: target professors, co-author papers, integrate into curricula, and build a citation flywheel that makes CarbonLens the de facto tool for CCUS research. The grad student who uses CarbonLens in their thesis is the Professional-tier customer in 3 years.

Phase 1 — Seed Partners (Months 1–6)

Target 3 specific professors at CCUS-active universities. Offer free Studio access in exchange for a published case study using CarbonLens. Target institutions: UTP (direct access via supervisor network), University of Wyoming School of Energy Resources (largest US-based CCS research programme), UT Austin Bureau of Economic Geology (Permian Basin CCS work). Approach: personalised email from Daniel T. Olagunju directly referencing shared publications in the CCUS space.

Phase 2 — Co-Authored Validation Paper (Months 6–12)

Co-author a paper specifically validating CarbonLens against published Sleipner monitoring data: “*CarbonLens Storage Studio: A Browser-Based CO₂ Storage Simulation Platform — Validation Against Sleipner and Decatur Field Data.*” Target: SPE Journal or IJGGC. Every CCUS researcher who reads this paper is a potential user. The paper is the product's most powerful marketing asset.

Phase 3 — Curriculum Integration (Months 12–24)

Work with 1–2 universities to integrate CarbonLens into an existing or new course on “Reservoir Engineering for Carbon Storage.” UTP is the natural first target given existing supervisor relationships. University of Wyoming is the most credible US target. Course integration provides: (1) sustained user base, (2) student feedback for product improvement, (3) academic legitimacy that drives consultancy sales, (4) potential co-development of teaching materials that establish CarbonLens as the standard tool for CCUS education.

10 — Full Financial Model & Long-Term Vision

\$10K

MRR target — Month 13 (PoC validated)

\$100K

MRR target — Month 24 (Full platform live)

\$1M+

ARR target — Month 36 (Platform market leader)

SCENARIO	YEAR 1 ARR	YEAR 2 ARR	YEAR 3 ARR	EXIT / PATH
Conservative	\$45K	\$180K	\$540K	Profitable bootstrap; raise Series A at \$1M ARR
Base Case	\$70K	\$480K	\$1.4M	Pre-seed at Month 14; Series A at Month 30
Optimistic	\$120K	\$900K	\$3.6M	Strategic acquisition by CMG/SLB at \$8–15M or raise growth round

Team & Hiring Plan (From Pre-Seed Proceeds)

ROLE	HIRE TIMING	COST	WHAT THEY BUILD
Full-Stack Engineer (React + Three.js)	Month 14–16	\$100– 130K	Module 6 (Capture/Transport/Utilization), Module 7 (MMV), GIS import
ML Engineer (Python, GMDH/GP)	Month 18–20	\$110– 140K	MMP model, impurity effects extension, Bayesian UQ upgrade, AI assistant
CCUS Domain Expert (PE / Geoscientist)	Month 20–24	\$100– 120K	Regulatory template accuracy, enterprise customer support, academic partnerships

ACQUISITION VS. INDEPENDENCE DECISION POINT

CMG launched Focus CCS in 2024. SLB is building CCS software capability. A productised, validated CarbonLens at \$1M ARR with a growing patent portfolio and 100+ paying customers is a \$5–15M acquisition target for either company. The decision to sell or raise and grow independently should be made with clear criteria: acquire at <\$5M ARR if the offer values the IP and team appropriately; raise independently if \$3M+ ARR is within 18 months. Do not accept an offer that undervalues the patent portfolio or removes the founder from the technical roadmap.